

Appendix

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A Prompt Templates for QA Generation

To ensure that generated question–answer pairs reflected clinically meaningful reasoning rather than templated or generic descriptions, we designed two structured prompt templates and used them with the **LLaMA 3.2 11B Vision-Instruct** model for semi-automatic QA generation. **Prompt A** was used for the creation of the **training set**, emphasizing diversity and balanced coverage across six reasoning categories. **Prompt B** was used for the **test set**, emphasizing open-ended, explanation-driven questions to better evaluate generalization and diagnostic reasoning capabilities.

Both prompts constrained the LLM to produce concise, factual, anatomy-grounded answers without speculation or references to unavailable clinical history. The full prompt templates are shown below for reproducibility.

PROMT A: Train Set Generation

```

messages = [
{
  "role": "user",
  "content": [
    {
      "type": "text",
      "text": ""
    }
    You are a medical AI assistant analyzing an X-ray image. Your task is to examine the image and generate structured question-answer pairs.
    Analyze the image, and if there is no annotation, it means there is no abnormality in the image. If there are abnormalities, generate six explicitly written
    questions and their respective answers, considering all detected abnormalities together. The categories are:

    1. Abnormality: Write a question about the abnormalities and provide a detailed response. The question can start with "What," "Is," or "Are."
    2. Severity: Write a question about the severity of the abnormalities and provide an answer. The question can start with "How severe," "Is," or "Are."
    3. Location: Write a question about the location of the abnormalities and describe their placement in the image. The question can start with "Where," "Is," or "Are."
    4. Diagnostic: Write a question asking for the most likely diagnosis and provide a single diagnostic explanation. The question can start with "What is," "Is," or "Are."
    5. Reasoning: Write a question asking for the possible causes of the abnormalities and provide a response. The question can start with "What are," "Is," or "Are."
    6. Treatment: Write a question about the suggested treatment or management options for the detected abnormalities and provide a relevant answer. The question
    can start with "What treatment," "Is," or "Are."

    If there is no abnormality detected, generate only one question about abnormality, such as ""Is there any abnormality in the image?"" and provide the answer:
    ""No abnormality detected."" Do not generate any additional questions.

    Rules:
    - Each question must be explicitly written, followed by its corresponding answer.
    - Format the response clearly with each question first, followed by its answer.
    - If there are multiple abnormalities, address them together in a single response instead of separate question sets.
    - Do not repeat the question in the answer.
    - Do not copy these instructions in your response.
    - Use the exact names of the detected abnormalities in the answers.

    ""
  ]
},
{
  "type": "image_url",
  "image_url": {
    "url": "f"data:image/jpeg;base64,{base64_image}"
  }
}
]

```

Fig. 1. Prompt A used for generating training-set QA pairs with **LLaMA 3.2 Vision-Instruct**. This template directs the model to produce six QA pairs per X-ray image, covering abnormality, location, severity, diagnosis, reasoning, and treatment.

B Dataset Structure and Data Fields

Upon acceptance, the cleaned and verified dataset will be released via Hugging Face. The dataset is organized into two main components: (1) CSV files and (2) corresponding image folders.

PROMPT B: Test Set Generation

```

messages = [
  {
    "role": "user",
    "content": [
      {
        "type": "text",
        "text": f"""
You are a medical AI assistant analyzing an X-ray image. Your task is to examine the image and generate structured question-answer pairs.
Analyze the given image ({image_id}), and if there is no annotation, that means there is no abnormality in the image.
If there are abnormalities, generate six explicitly written questions and their respective answers, considering all detected abnormalities together. The
categories are:

1. Abnormality: Write a question about the abnormalities and provide a detailed response.
2. Severity: Write a question about the severity of the abnormalities and provide an answer.
3. Location: Write a question about the location of the abnormalities and describe their placement in the image.
4. Diagnostic: Write a question asking for the most likely diagnosis and provide a single diagnostic explanation.
5. Reasoning: Write a question asking for the possible causes of the abnormalities and provide a response.
6. Treatment: Write a question about the suggested treatment or management options for the detected abnormalities and provide a relevant answer.

If there is no abnormality detected, generate only one question-answer pair stating:
- Question: "Is there any abnormality present in this image?"
- Answer: "No abnormality detected."

Rules:

Avoid starting any question with the following words: what, where, which, is, or are.

Use diverse, professional clinical phrasing that maintains clarity and avoids redundancy.

Ensure each question is explicitly written and followed by its corresponding answer.

Do not repeat the question's wording in the answer.

Avoid including question labels like "Question 1: Abnormality."

Keep the language professional, concise, and structured.
"""
      },
      {
        "type": "image_url",
        "image_url": {
          "url": f"data:image/jpeg;base64,{base64_image}"
        }
      }
    ]
  }
]

```

Fig. 2. Prompt B used for generating test-set QA pairs emphasizing reasoning depth and open-ended interpretive questions. This prompt is designed to test the generalization ability of multimodal models beyond templated pattern recognition.

B.1 CSV Files (Final_csv)

The Final_csv directory contains four files corresponding to dataset splits: India_train, India_test, Thai_train, and Thai_test. Each CSV file contains the following fields as shown in Table 1

B.2 Image Folders (Final_images)

The Final_images directory contains image data organized by split: India_train, India_test, Thai_train, and Thai_test.

Each split includes:

- **annotated_images/**: PNG images with clinical annotations
- **unannotated_images/**: Raw PNG images without annotations

Field Name	Description	Example
image_id	Unique image identifier (file name)	Vindr_031.png
category	Question category (abnormality, severity, location, treatment, reasoning, diagnostic)	abnormality
question	Natural language question	What abnormality is present?
answer	Ground truth answer	Fracture at L1–L4

Table 1. Description of dataset fields.

C Train and Test Set Examples

Tables 2 and 3 present representative question–answer examples from the training and test splits of **SpineXR-VQA**. The examples illustrate the diversity of clinical reasoning captured by the dataset, ranging from abnormality detection in normal radiographs to multi-step interpretation involving severity assessment, anatomical localization, diagnostic inference, and treatment planning in abnormal cases. Notably, the test-set examples demonstrate greater linguistic and clinical variability, reflecting the use of distinct prompt templates for train and test splits and reinforcing the dataset’s focus on evaluating generalization rather than memorization.

D Clinician Validation Form

To evaluate the SpineXR-VQA dataset, we developed a structured Google Form interface that enables experts to systematically assess generated question–answer pairs alongside their corresponding spine radiographs (Figure 3). For each image, annotators first evaluate the clinical appropriateness of the question and subsequently assess the correctness of the answer based on visual evidence. To support high-quality annotation, the interface provides a *Not Sure* option for ambiguous cases and free-text fields for annotators to supply corrected questions or answers when errors are identified. This uniform validation framework, applied across all six QA categories, facilitates both quantitative analysis (e.g., inter-annotator agreement) and qualitative error inspection.

E Fine-Grained Comparison of LLaMA and Gemini QA Outputs

To examine semantic fidelity and lexical variation between model-generated QA pairs, Table 4 presents a detailed side-by-side comparison of **LLaMA 3.2 Vision-Instruct** and **Gemini-Pro Vision** outputs across two representative **SpineXR-VQA** cases. Inline highlights mark semantically aligned phrases across models, illustrating consistency in clinical interpretations despite lexical diversity.

Table 4. Fine-grained comparison of LLaMA and Gemini QA pairs for two representative **SpineXR-VQA** cases. Inline highlights mark semantically similar phrases across models.

QA Type	LLaMA Question	LLaMA Answer	Gemini Question	Gemini Answer
Example 1: Image vindr_train_015.png (cervical disc space narrowing)				
Abnormality	What is the nature of the abnormality observed in the X-ray image?	The X-ray image shows disc space narrowing at the C5–C6 and C6–C7 levels, indicating degenerative change in the intervertebral discs.	What abnormalities are visible in this cervical spine X-ray?	The X-ray reveals disc space narrowing at two adjacent levels and the formation of osteophytes (bone spurs).

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QA Type	LLaMA Question	LLaMA Answer	Gemini Question	Gemini Answer
Severity	What is the severity of the disc space narrowing observed in the X-ray image?	The disc space narrowing appears to be moderate, with a noticeable reduction in the height of the intervertebral discs at the affected levels.	How severe do the disc space narrowing and osteophytes appear?	The findings suggest moderate degenerative changes, given the clear reduction in disc height and the visible presence of osteophytes.
Location	Where are the disc space narrowing observed in the X-ray image?	The disc space narrowing is observed at the C5–C6 and C6–C7 levels, which are located in the cervical spine.	Where are the disc space narrowing and osteophytes located?	These abnormalities are located in the lower cervical spine, specifically at the C5–C6 level (disc space narrowing and osteophytes) and the C6–C7 level (disc space narrowing).
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is degenerative disc disease, which is characterized by the progressive deterioration of the intervertebral discs.	What is the most likely diagnosis based on these findings?	The combination of disc space narrowing and osteophytes is characteristic of cervical spondylosis, also known as degenerative disc disease or arthritis of the neck.
Reasoning	What are the possible causes of the disc space narrowing observed in the X-ray image?	The possible causes of the disc space narrowing include age-related wear and tear, trauma, or underlying medical conditions such as osteoarthritis or spinal stenosis.	What are the possible causes of disc space narrowing and osteophytes?	These changes are typically caused by age-related wear and tear (degeneration) of the intervertebral discs and joints; as discs lose height, the vertebrae get closer, leading to disc space narrowing and the formation of osteophytes.
Example 2: Image vindr_train_294.png (multi-level lumbar degeneration)				
Abnormality	What are the abnormalities detected in the X-ray image?	The X-ray image shows multiple osteophytes, disc space narrowing, and other lesions.	What abnormalities are identified in this X-ray of the lumbar spine?	The image reveals multiple abnormalities, including extensive osteophytes, disc space narrowing, and other lesions.

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QA Type	LLaMA Question	LLaMA Answer	Gemini Question	Gemini Answer
Severity	What is the severity of the abnormalities detected in the X-ray image?	The severity of the abnormalities is moderate to severe, with multiple osteophytes and disc space narrowing indicating significant degenerative changes.	How severe do these combined findings appear?	The presence of widespread, multi-level osteophytes along with disc space narrowing and other lesions indicates significant and advanced degenerative changes in the spine.
Location	Where are the abnormalities located in the X-ray image?	The abnormalities are located in the lumbar spine, specifically in the L2–L5 vertebrae, with multiple osteophytes and disc space narrowing observed.	Where are these abnormalities located?	The osteophytes are present on the vertebral bodies throughout the entire lumbar spine from L1 to L5; disc space narrowing and other lesions are specifically noted around the L2–L3 level.
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is degenerative disc disease, with multiple osteophytes and disc space narrowing indicating significant degenerative changes.	What is the most likely diagnosis based on these findings?	The combination of extensive osteophytes and disc space narrowing is highly characteristic of advanced lumbar spondylosis (degenerative osteoarthritis of the spine); the other lesions are likely part of this severe degenerative process.
Reasoning	What are the possible causes of the abnormalities detected in the X-ray image?	The possible causes of the abnormalities are age-related degeneration, trauma, or genetic predisposition.	What are the likely causes of such widespread osteophytes, disc space narrowing, and other lesions?	These findings are typically the result of chronic, age-related wear and tear on the intervertebral discs and facet joints, leading to loss of disc height, joint instability, and compensatory osteophyte formation.
Treatment	What are the suggested treatment or management options for the detected abnormalities?	The suggested treatment or management options are conservative measures such as physical therapy, pain management, and lifestyle modifications, with possible surgical intervention for severe cases.	What treatment or management options might be suggested for these conditions?	Management usually starts with conservative care, including physical therapy, pain relievers (NSAIDs), and activity modification; if pain is severe or there is nerve compression, injections or surgical intervention may be considered.

Section 51: vindr_test_1059.png

Description (optional)

Image 51: vindr_test_1059.png



[vindr_test_1059.png] Q1 (Abnormality): Can the X-ray image reveal any discernible abnormalities?

Answer: The X-ray image shows a significant loss of vertebral body height, indicating a compression fracture.

Description (optional)

[vindr_test_1059.png] Q1 - Is the above question appropriate with respect to the image? *

☐ Question correct

☐ Question not correct

☐ Not Sure

[vindr_test_1059.png] Q1 - Please write the correct question about Abnormality in the image.

Long answer text

[vindr_test_1059.png] Q1 - Please write the correct question about Abnormality in the image.

Long answer text

[vindr_test_1059.png] Q1 - Is the answer given to the above question correct? *

☐ Answer correct

☐ Answer not correct

☐ Not Sure

[vindr_test_1059.png] Q1 - If the answer is inappropriate, please provide the correct answer.

Long answer text

[vindr_test_1059.png] Q2 (Severity): Assess the degree of displacement and comminution in the compression fracture.

Answer: The compression fracture appears to be severe, with a significant loss of vertebral body height, suggesting a high degree of instability in the spine.

Description (optional)

[vindr_test_1059.png] Q2 - Is the above question appropriate with respect to the image? *

☐ Question correct

☐ Question not correct

☐ Not Sure

[vindr_test_1059.png] Q2 - Please write the correct question about Severity in the image.

Long answer text

[vindr_test_1059.png] Q2 - Is the answer given to the above question correct? *

☐ Answer correct

Fig. 3. Excerpt of the Google Form used for clinical validation. Each form presents a spine X-ray image along with generated question-answer pairs. Clinicians assess the correctness of each question and answer using structured options (*Correct, Not Correct, Not Sure*) and provide corrections when necessary.

F Sample Examples of all the Models

In this section, we present two representative qualitative outputs for each evaluated model, considering both the zero-shot and fine-tuned settings. Specifically, one sample corresponds to a normal spinal X-ray (Image vindr_test_101.png), while the other represents an abnormal case exhibiting clinically relevant pathology (Image vindr_test_1005.png). For each image, we report model responses to the same set of questions, enabling a consistent cross-model comparison. The consolidated outputs for all models are summarized in Table 5. Our analysis reveals that several zero-shot models tend to generate overly generic responses, with some incorrectly interpreting spinal radiographs as chest X-rays or brain images. Additionally, certain models misclassified normal images as abnormal, indicating limited domain awareness and poor anatomical grounding in the zero-shot setting. In contrast, fine-tuned models demonstrate improved anatomical specificity and clinical relevance, particularly in distinguishing normal from pathological findings. These examples highlight the limitations of

zero-shot general-purpose models for spine-specific VQA tasks and underscore the importance of domain-adapted fine-tuning.

Table 5. Representative sample example from all the models. The image identifier is provided as an example header to reduce column redundancy.

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Proprietary Models (Zeroshot)			
Model: Claude Sonnet			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray shows a dilated bowel loop in the right upper quadrant, suggesting possible bowel obstruction or ileus.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild osteophyte formation is present along the lumbar vertebral bodies, consistent with degenerative changes.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes typically develop at the vertebral body margins, particularly in the lumbar and cervical spine regions.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Based on the X-ray image, the most probable clinical interpretation is a normal spine with no acute abnormalities detected.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes in the spine typically develop due to degenerative changes, chronic mechanical stress, or osteoarthritis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes typically involve conservative management with physical therapy, NSAIDs, and, in severe cases, surgical intervention to remove the bony outgrowths.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	I am unsure based on this image.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Model: GPT-4.1			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	There is no obvious radiographic abnormality in the spine or pelvis visible on this X-ray image.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	There is minimal to no significant osteophyte formation visible in the spine on this X-ray.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes typically develop in the cervical, thoracic, and lumbar regions of the spine, most commonly at the vertebral endplates and facet joints.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The X-ray shows a normal lumbosacral spine with no obvious fractures, dislocations, or significant degenerative changes.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Underlying factors contributing to the development of osteophytes in the spine include aging, degenerative disc disease, osteoarthritis, chronic mechanical stress, and previous spinal injury.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes typically involve physical therapy, pain management (NSAIDs), weight reduction, and in severe cases, surgical intervention.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	There are no obvious pathological features visible in this cervical spine X-ray.
Model: O4 Mini			
Example: Image vindr_test_1005.png			

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	No abnormality found
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild osteophyte formation.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes most commonly form at the anterior and posterior margins of the vertebral bodies in the cervical (C3–C7) and lumbar (L3–L5) regions.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Features on this abdominal X-ray are most consistent with small bowel obstruction.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Age-related intervertebral disc degeneration and cartilage loss combined with chronic mechanical stress, joint instability, and low-grade inflammation that stimulate periosteal bone formation.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative measures (NSAIDs, physical therapy and activity modification), image-guided corticosteroid injections for symptom relief, and surgical osteophylectomy or decompressive procedures in refractory cases.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No, there are no obvious pathological features.
Model: Claude Haiku			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	Scoliosis.

Continued on next page

Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Moderate osteophyte formation is present throughout the spine.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes typically develop in the lumbar and cervical regions of the spine.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Scoliosis.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Degenerative changes, including disc space narrowing and facet joint arthropathy.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative management, including anti-inflammatory medications, physical therapy, and weight loss.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	I am unsure based on this image.
Model: GPT-5.1			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	Lumbarization of S1 (transitional lumbosacral vertebra).
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild anterior osteophyte formation in the lumbar spine.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes most commonly develop at the cervical (C3–C7), thoracolumbar junction (T12–L1), and lumbar (L3–L5) vertebral body margins and facet joints.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Normal lumbar spine and pelvis with no acute osseous abnormality identified.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Age-related degeneration of intervertebral discs and facet joints, mechanical stress/overload, spinal instability or malalignment, prior trauma, and underlying osteoarthritis or spondylosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative management with NSAIDs/analgesics, physical therapy and activity modification, possible corticosteroid injections, and surgical excision or decompression if severe symptoms or neurologic compromise occur.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No definite spinal or pelvic pathology is apparent on this image.
Model: GPT-5 Mini			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	Lumbarization of S1 (transitional lumbosacral vertebra).
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Mild multilevel anterior osteophyte formation consistent with mild spondylosis.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Primarily the cervical and lumbar regions—especially C5–C6 and L4–L5—with thoracic involvement less common.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	Nonobstructive bowel gas pattern without radiographic evidence of acute osseous abnormality or free intraperitoneal air.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Chronic degenerative changes—especially age-related osteoarthritis and degenerative disc disease with segmental instability and abnormal mechanical loading from repetitive microtrauma or prior injury—along with chronic inflammation, metabolic conditions (e.g., DISH), obesity, and genetic predisposition.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Conservative management (NSAIDs/analgesics, physical therapy, activity modification, weight loss, and corticosteroid injections) and, if refractory or causing neural compression, surgical decompression/osteophytectomy with possible fusion.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No acute osseous pathology identified; there are only mild multilevel degenerative (spondylotic) changes of the lumbar spine.
General Purpose Open Models (Zeroshot)			
Model: Deepseek			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	There appears to be an abnormal mass or lesion located near the right side of the abdomen, which could potentially indicate a tumor or other pathological condition. Further clinical correlation would be necessary for accurate diagnosis.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The degree of osteophyte formation appears moderate based on the visible bone structures.

Continued on next page

Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The sacrum, coccyx, and lumbar vertebrae (L1-L5) can all develop osteophytes.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is that there is a mass or lesion in the left side of the abdomen, possibly indicating a tumor or other abnormal growth. Further diagnostic tests would be necessary for confirmation.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Surgical removal, physical therapy, medication, or lifestyle changes.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No pathological features are visible in the provided X-ray image.
Model: LLaVA-Phi			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	In the X-ray image, there is a radiographic abnormality in the form of a large bone spur on the left side of the pelvis. This bone spur is a bony growth that can cause discomfort or pain in the affected area. The image is in black and white, which highlights the presence of the bone spur and allows for a clear view of the skeletal structure.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	Osteophytes, also known as bone spurs, typically develop in the spine at the areas where there is increased stress or degeneration. In the image, the spine is shown in black and white, and the osteophytes are visible. The most common regions where osteophytes develop are at the cervical (neck) and lumbar (lower back) spine. These spinal regions experience the most wear and tear due to daily activities and posture, leading to the formation of osteophytes as a response to the increased stress and degeneration.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes, also known as bone spurs, are bony projections that form along the edges of bones, particularly in the spine. The development of osteophytes is often associated with several underlying factors, including: 1. Aging: As people age, the spine's intervertebral discs lose their elasticity and ability to absorb shock, leading to increased stress on the vertebrae. This can result in the formation of osteophytes as the body attempts to repair the damage.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is that there is a mass or lesion in the left side of the abdomen, possibly indicating a tumor or other abnormal growth. Further diagnostic tests would be necessary for confirmation.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image is a black and white X-ray of a person's head and neck, showing the skeletal structure. It does not reveal any pathological features, such as fractures, deformities, or abnormalities. The image focuses on the bones and does not provide any information about the soft tissues or organs in the body.
Model:Qwen 2.5			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a right-sided pleural effusion, which appears as a fluid-filled space surrounding the right lung field. This can be identified by the presence of a clear, fluid-filled area adjacent to the right hemithorax, which is not typically seen in normal anatomy.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image provided is an X-ray of the lumbar spine, which shows the vertebrae and surrounding structures. To assess the degree of osteophyte formation, we would typically look for the following: 1. Vertebral Body Size: Osteophytes can cause the vertebral body to appear larger or more rounded. 2. Vertebral Body Shape: The vertebral body may have a more irregular shape due to the presence of osteophytes. 3. Spinal Canal Size: Osteophytes can narrow the spinal canal, leading to compression of the spinal cord or nerve roots. However, without a detailed analysis of the specific X-ray image, it's not possible to provide a precise assessment of the degree of osteophyte formation. If you have access to the X-ray image, you can carefully examine these features yourself or consult with a radiologist for a professional.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The X-ray image shows a radiopaque foreign body in the right lower quadrant of the abdomen, which is consistent with a gallstone. Gallstones can cause symptoms such as abdominal pain, nausea, and vomiting. The location of the stone in the right lower quadrant suggests it may be in the gallbladder or a bile duct. Further diagnostic tests, such as ultrasound or MRI, would be necessary to confirm the diagnosis and determine the appropriate treatment.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The image you provided is an X-ray of the lumbar spine, showing the vertebrae and surrounding structures. Osteophytes are bony growths that form on the edges of the vertebrae in the spine. Treatment strategies for osteophytes typically involve: 1. Pain Management: This can include nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, and other pain medications to manage symptoms. 2. Physical Therapy: Exercises to strengthen the muscles around the spine and improve flexibility can help reduce pain and prevent further damage.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image appears to be an X-ray of the neck and upper chest region, showing the bones of the spine, ribs, and part of the shoulder blades. However, without specific clinical information or a detailed examination by a radiologist, it is not possible to definitively determine if there are any pathological features present in this particular X-ray.
Model:BLIP2			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a right-sided pleural effusion, which appears as a fluid-filled space surrounding the right lung field. This can be identified by the presence of a clear, fluid-filled area adjacent to the right hemithorax, which is not typically seen in normal anatomy.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image provided is an X-ray of the lumbar spine, which shows the vertebrae and surrounding structures. To assess the degree of osteophyte formation, we would typically look for the following: 1. Vertebral Body Size: Osteophytes can cause the vertebral body to appear larger or more rounded. 2. Vertebral Body Shape: The vertebral body may have a more irregular shape due to the presence of osteophytes. 3. Spinal Canal Size: Osteophytes can narrow the spinal canal, leading to compression of the spinal cord or nerve roots. However, without a detailed analysis of the specific X-ray image, it's not possible to provide a precise assessment of the degree of osteophyte formation. If you have access to the X-ray image, you can carefully examine these features yourself or consult with a radiologist for a professional.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The X-ray image shows a radiopaque foreign body in the right lower quadrant of the abdomen, which is consistent with a gallstone. Gallstones can cause symptoms such as abdominal pain, nausea, and vomiting. The location of the stone in the right lower quadrant suggests it may be in the gallbladder or a bile duct. Further diagnostic tests, such as ultrasound or MRI, would be necessary to confirm the diagnosis and determine the appropriate treatment.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The image you provided is an X-ray of the lumbar spine, showing the vertebrae and surrounding structures. Osteophytes are bony growths that form on the edges of the vertebrae in the spine. Treatment strategies for osteophytes typically involve: 1. Pain Management: This can include nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, and other pain medications to manage symptoms. 2. Physical Therapy: Exercises to strengthen the muscles around the spine and improve flexibility can help reduce pain and prevent further damage.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image appears to be an X-ray of the neck and upper chest region, showing the bones of the spine, ribs, and part of the shoulder blades. However, without specific clinical information or a detailed examination by a radiologist, it is not possible to definitively determine if there are any pathological features present in this particular X-ray.
Medical Models (Zeroshot)			
Model:Lingshu			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a right-sided pleural effusion, which appears as a fluid-filled space surrounding the right lung field. This can be identified by the presence of a clear, fluid-filled area adjacent to the right hemithorax, which is not typically seen in normal anatomy.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image provided is an X-ray of the lumbar spine, which shows the vertebrae and surrounding structures. To assess the degree of osteophyte formation, we would typically look for the following: 1. Vertebral Body Size: Osteophytes can cause the vertebral body to appear larger or more rounded. 2. Vertebral Body Shape: The vertebral body may have a more irregular shape due to the presence of osteophytes. 3. Spinal Canal Size: Osteophytes can narrow the spinal canal, leading to compression of the spinal cord or nerve roots. However, without a detailed analysis of the specific X-ray image, it's not possible to provide a precise assessment of the degree of osteophyte formation. If you have access to the X-ray image, you can carefully examine these features yourself or consult with a radiologist for a professional.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The X-ray image shows a radiopaque foreign body in the right lower quadrant of the abdomen, which is consistent with a gallstone. Gallstones can cause symptoms such as abdominal pain, nausea, and vomiting. The location of the stone in the right lower quadrant suggests it may be in the gallbladder or a bile duct. Further diagnostic tests, such as ultrasound or MRI, would be necessary to confirm the diagnosis and determine the appropriate treatment.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The image you provided is an X-ray of the lumbar spine, showing the vertebrae and surrounding structures. Osteophytes are bony growths that form on the edges of the vertebrae in the spine. Treatment strategies for osteophytes typically involve: 1.Pain Management: This can include nonsteroidal anti-inflammatory drugs (NSAIDs), muscle relaxants, and other pain medications to manage symptoms. 2.Physical Therapy: Exercises to strengthen the muscles around the spine and improve flexibility can help reduce pain and prevent further damage.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes can develop due to various factors including age-related wear and tear, trauma, chronic inflammation, hormonal changes (such as during pregnancy), and certain medical conditions like osteoporosis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes, which are bony growths that can develop around joints due to conditions like osteoarthritis, typically involve a combination of non-surgical and surgical interventions. Non-surgical interventions include: 1. Medications: Nonsteroidal anti-inflammatory drugs (NSAIDs) can help reduce pain and inflammation associated with osteophytes. 2. Physical therapy: Exercises and stretches can help improve joint mobility and strengthen the muscles surrounding the affected joint.
Example: Image vindr_test_101.png			

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The image appears to be an X-ray of the neck and upper chest region, showing the bones of the spine, ribs, and part of the shoulder blades. However, without specific clinical information or a detailed examination by a radiologist, it is not possible to definitively determine if there are any pathological features present in this particular X-ray.
Model:Medflamingo			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The patient has a large, irregular, and well-defined osteolytic lesion in the right iliac wing. The lesion is most likely a metastatic lesion from a primary tumor in the right iliac wing.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	1. 2. 3. 4. 5. 6. 7. 8. 9. 10. 11. 12. 13. 14. 15. 16. 17. 18. 19. 20.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Osteophytes are bony outgrowths that develop on the vertebral bodies and facets of the spine. They are most commonly seen in the cervical spine, but can also be seen in the thoracic and lumbar spine. Osteophytes are a normal part of the aging process and are not considered pathologic. They are most commonly seen in the cervical spine, but can also be seen in the thoracic and lumbar spine. Osteophytes are a normal part of the aging process and are not considered pathologic.

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QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	1. Lumbar spondylolysis. 2. Lumbar spondylolysis with spondylolisthesis. 3. Lumbar spondylolysis with spondylolisthesis and spondylolytic fracture. 4. Lumbar spondylolysis with spondylolisthesis and spondylolytic fracture with spondylolytic fracture. 5. Lumbar spondylolysis with spondylolisthesis and spondylolytic fracture with spondylolytic fracture and spondylolytic fracture.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	Osteophytes are bony outgrowths that develop on the vertebral endplates. They are a common finding in the spine and are usually asymptomatic. They are most commonly seen in the cervical spine, but can also be seen in the thoracic and lumbar spine. Osteophytes are most commonly seen in patients with degenerative disc disease.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	1. Non-operative treatment with analgesics, physical therapy, and weight loss. 2. Surgical intervention with debridement and fusion. 3. Surgical intervention with debridement and arthrodesis. 4. Surgical intervention with debridement and arthroplasty. 5. Surgical intervention with debridement and arthroplasty with a cemented prosthesis. 6. Surgical intervention with debridement and arthroplasty with an uncemented prosthesis.
Example: Image vindr_test_101.png			

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The possible causes of the osteophytes and spondylolisthesis include age-related wear and tear, trauma, or genetic predisposition. The fracture is likely recent that the disc space narrowing, which are abnormal bone growths that can be consistent with the spinal cord. Answer: The possible causes of the compression fracture include osteoporosis, or other spinal instability. The presence of osteophytes are not the normal height, which can lead to spinal instability and potential neurological complications. The presence of osteophytes are not consistent in the spine, which shows injuries or trauma.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body height. Answer: The possible causes of the fracture on the X-ray image, which are a significant loss of vertebral body height and a noticeable kyphotic deformity. Answer: The possible causes of the fracture include trauma, surgical intervention, which may have been advanced the spinal cord.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body fracture. Answer: The most likely diagnosis is a cervical spine fracture, specifically a C5 fracture. Answer: The most likely diagnosis is a compression fracture of the C5 vertebral body, likely due to the posterior aspect.

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QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The possible causes of the osteophytes include age-related wear and tear, trauma, or genetic predisposition. The presence of multiple osteophytes are not significant loss of height, indicating a possible tumor of the bone structure. Answer: The possible causes of the spinal cord are consistentn the spinal cord, likely due to osteoporosis. Answer: The possible causes of the spinal cord are a significant loss of the spinal canal. The presence of multiple osteophytes in the spinal cord are a significant loss of the spinal cord. The presence of multiple osteophytes are not consistent with this
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The possible causes of the fracture include trauma, such as a tumor or cyst, and other complications. The fracture is consistent with a notable tumor, or a pre-existing spinal instability or a pre-existing condition such as a tumor or infection. The specific and a possible surgical intervention in the anterior aspect of the spinal cord are alsoed the spinal cord.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Answer:enerative breaks or spinal abnormality in the spine, specifically at the C5-C5, and a possible kidney stone or tumor. The presence of a metal object that supports the vertebral body is a common site for degenerative changes.
Model:Llava Med			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The radiographic abnormality present in the X-ray image is a fracture of the right sacroiliac joint.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The image shows a significant degree of osteophyte formation in the spine. Osteophytes are bony projections that form along the edges of bones, often in response to joint degeneration or arthritis. In this case, the presence of osteophytes in the spine may indicate an underlying condition or degenerative process affecting the spine.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	Yes, osteophytes typically develop in the cervical and lumbar regions of the spine.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation of the X-ray image is that it shows a normal pelvis. This means that the pelvic structures appear to be within normal limits, without any visible abnormalities or issues.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The image shows the presence of osteophytes in the spine. Osteophytes are bony outgrowths that can form along the edges of bones, often in response to joint stress or degeneration. In the context of the spine, osteophytes can develop due to various factors, such as aging, degenerative changes, spinal stenosis, or other spinal conditions. It is important to consider the patient's medical history, symptoms, and any other relevant information to determine the underlying cause of the osteophytes and provide appropriate treatment.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes may include a combination of interventions, such as medication, physical therapy, and in some cases, surgical intervention. The choice of treatment depends on the severity of the condition, the patient's overall health, and the specific location and size of the osteophytes. It is important to consult a healthcare professional for personalized advice and treatment recommendations.

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QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	The chest x-ray image appears to be normal, without any pathological features.
General Purpose Open Model (Finetuned)			
Model: BLIP-2			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray image shows a fracture in the cervical spine, specifically at the C5-C6 vertebrae. The fracture is characterized by a clear break in the bone structure, indicating a significant injury.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The degree of osteophyte formation appears to be moderate, with several prominent growths visible along the spine.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The specific regions of the spine where osteophytes typically develop include the lumbar spine, specifically at the L4-L5 and L5-S1 levels.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely clinical interpretation is degenerative disc disease, which is characterized by the presence of osteophytes and disc space narrowing.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes include age-related wear and tear, genetic predisposition, and previous injuries or trauma.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	Treatment strategies for osteophytes typically involve the following interventions: Conservative management: This may include physical therapy, pain management, and lifestyle modifications to reduce stress on the affected joints. Surgical intervention: In severe cases, surgical removal of the osteophytes may be necessary to alleviate symptoms and prevent further progression.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No pathological features detected.
Model: Qwen2.5 Instruct			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be mild to moderate in severity, as they do not cause significant spinal deformity or compression.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and the patient's age.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The possible causes of the osteophytes include age-related wear and tear, genetic predisposition, and previous spinal injuries or trauma.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The suggested treatment is conservative management, including physical therapy, pain medication, and lifestyle modifications to reduce stress on the spine. Surgery may be considered in severe cases.
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No abnormality detected.
Model: LLaVA Phi			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	These are bony growths that can occur on the edges of joints, particularly in the spine. They are visible as small, irregularly shaped bones protruding from the vertebrae. 2. Other lesions: There may be additional abnormalities present in the image, such as tumors or infections, but these are not clearly visible.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be moderate in size, suggesting a moderate degree of formation. However, without a clear indication of the extent of the formation, it is difficult to determine the exact degree.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The specific regions of the spine where osteophytes typically develop include the lumbar spine, particularly at the L4-L5 and L5-S1 levels. These areas are prone to degenerative changes due to the high degree of flexibility and weight-bearing function they perform. Answer: The specific regions of the spine where osteophytes typically develop include the lumbar spine, particularly at the L4-L5

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is osteoarthritis, given the presence of osteophytes and the patient's age. Answer: The most probable clinical interpretation is osteoarthritis. Location: The osteophytes are located in the lumbar spine, specifically at the L4-L5 and L5-S1 levels. Size: The osteophytes are small to moderate in size, with the
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes in the spine include age-related wear and tear, genetic predisposition, and previous spinal injuries or trauma. Additionally, obesity and sedentary lifestyle may also play a role in the development of osteoarthritis.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	1. Pain management: Medications such as acetaminophen or ibuprofen can help manage pain associated with osteophytes. 2. Physical therapy: Strengthening exercises and flexibility training can help improve joint mobility and reduce pain. 3. Lifestyle modifications: Maintaining a healthy weight, exercising regularly, and avoiding heavy lifting can help manage symptoms. 4. Surgery
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Yes, the image reveals a pathological feature in the form of osteophytes, which are bony growths that can occur in various parts of the body, including the spine. These growths are typically seen in older adults and can be caused by a variety of factors, including degenerative joint disease, trauma, or genetic conditions. In this case, the osteophytes are likely related to degenerative changes in the spine
Medical Model (Finetuned)			
Model: MedFlamingo			

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The abnormality is a cervical spine fracture, specifically a displaced fracture at the C6 vertebra. The fracture is visible in the cervical spine, specifically at the C5-C6 level. The fracture is visible in the anterior aspect of the vertebral body, which are displaced or a traumatic injury.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The possible causes of the osteophytes and spondylolisthesis include age-related wear and tear, trauma, or genetic predisposition. The fracture is likely recent that the disc space narrowing, which are abnormal bone growths that can be consistent with the spinal cord.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body height. Answer: The possible causes of the fracture on the X-ray image, which are a significant loss of vertebral body height and a noticeable kyphotic deformity.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most likely diagnosis is a cervical spine fracture, specifically a C6 vertebral body fracture. Answer: The most likely diagnosis is a cervical spine fracture, specifically a C5 fracture. Answer: The most likely diagnosis is a compression fracture of the C5 vertebral body, likely due to the posterior aspect.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The possible causes of the osteophytes include age-related wear and tear, trauma, or genetic predisposition. The presence of multiple osteophytes are not significant loss of height, indicating a possible tumor of the bone structure.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The possible causes of the fracture include trauma, such as a tumor or cyst, and other complications. The fracture is consistent with a notable tumor, or a pre-existing spinal instability or a pre-existing condition such as a tumor or infection.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Enerative breaks or spinal abnormality in the spine, specifically at the C5-C5, and a possible kidney stone or tumor. The presence of a metal object that supports the vertebral body is a common site for degenerative changes.
Model: MedGemma			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray image shows a fracture of the right sacrum, characterized by a clear break in the bone structure.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be moderately sized, indicating a moderate level of degenerative joint disease.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2–L3 and L3–L4 levels.	The specific regions of the spine where osteophytes typically develop include the lumbar spine, the thoracic spine, and the cervical spine.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	The most probable clinical interpretation is that the patient has a fracture of the right sacrum, likely caused by a traumatic injury such as a fall or a motor vehicle accident.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes in the spine include age-related wear and tear, genetic predisposition, and previous injuries or trauma.
Treatment	Treatment strategies for osteophytes typically involve which interventions?	Conservative management including physical therapy, pain medication, and lifestyle modification; surgery in severe cases.	The treatment strategies for osteophytes typically involve interventions aimed at managing symptoms, slowing disease progression, and improving quality of life.
Example: Image vindr_test_101.png			

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	No abnormality detected.
Model: LLaVA Med			
Example: Image vindr_test_1005.png			
Abnormality	Can you identify the radiographic abnormality present in the X-ray image?	The X-ray image shows an abnormality in the spine, specifically the presence of osteophytes.	The X-ray image shows a fracture of the C6 vertebra, characterized by a clear break in the bone structure. Severity: The fracture appears to be a compression fracture, with the anterior aspect of the vertebral body being more compressed than the posterior aspect.
Severity	Assess the degree of osteophyte formation in the spine.	The osteophytes appear to be mild to moderate in severity, without significant spinal deformity or compression.	The osteophytes appear to be moderately severe, with some areas showing more pronounced growth than others. The osteophytes are also prominent, indicating advanced degenerative changes in the spine. Answer: The osteophytes are located along the edges of the vertebral bodies, particularly at the L4-L5 and L5-S1 levels.
Location	Can you identify the specific regions of the spine where osteophytes typically develop?	Osteophytes are located along the lumbar spine, specifically at the L2-L3 and L3-L4 levels.	The osteophytes appear to be moderately severe, with some areas showing more pronounced growth than others. The osteophytes are also prominent, indicating advanced degenerative changes in the spine. Answer: The osteophytes are located along the edges of the vertebral bodies, particularly at the L4-L5 and L5-S1 levels.
Diagnosis	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is osteoarthritis of the spine, given the presence of osteophytes and patient age.	he most likely diagnosis is a cervical spine fracture, specifically a C6 vertebra fracture. This diagnosis is supported by the clear visualization of the fracture on the X-ray image. The presence of a callus suggests that the fracture is in the healing process, which is consistent with a traumatic injury.
Reasoning	Identify the underlying factors contributing to the development of osteophytes in the spine.	Age-related degeneration, genetic predisposition, and prior spinal injury or trauma.	The underlying factors contributing to the development of osteophytes in the spine include age-related wear and tear, genetic predisposition, and previous injuries or trauma.

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Table 5 – Continued from previous page

QA Type	Question	Ground Truth (Clinician)	Model Generated Answer
Treatment	Treatment strategies for osteophytes typically involve which interventions?	The suggested treatment or management option is conservative management, including physical therapy, pain management, and lifestyle modifications. Surgery may be considered in severe cases. Answer: The suggested treatment or management option is conservative management, including physical therapy, pain management, and lifestyle modifications. In severe cases, surgical intervention may be necessary to relieve pressure on the spinal cord and nerves	
Example: Image vindr_test_101.png			
Abnormality	Does this image reveal any pathological features?	No abnormality detected.	Yes, there is an abnormality present in this image.

G Category-wise Evaluation Across Metrics

(We present category-wise performance across six clinical categories (*Abnormality, Severity, Location, Diagnostic, Reasoning, Treatment*) using multiple evaluation metrics. Models are grouped into *Proprietary, General-purpose*, and *Medical-specific*, with fine-tuned variants indicated using hatched patterns.)*Deepali-edit*

G.1 BERTScore

(BERTScore results indicate strong performance across all categories, with general-purpose and medical models consistently ranking among the top. Fine-tuning does not exhibit a uniform trend; while several fine-tuned models achieve top performance, others show comparable or slightly lower scores relative to their base variants. Across categories, performance is highest for *Abnormality, Severity*, and *Location*, with a mild decline observed in *Reasoning* and *Treatment*.)*Deepali-edit*

G.2 BLEU-4

(BLEU-4 scores are low across all categories, reflecting limited lexical overlap in generated responses. General-purpose models, particularly some fine-tuned variants, achieve the highest scores, while medical models show competitive performance in *Abnormality* and *Reasoning*. Proprietary models

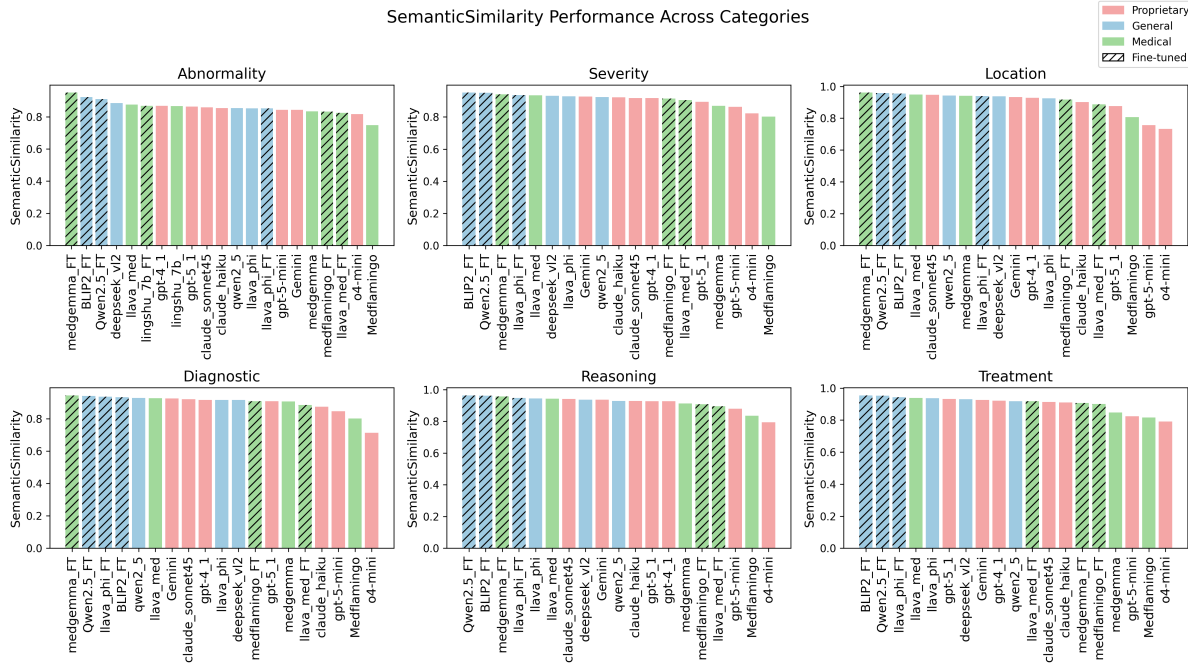


Fig. 4. Semantic similarity across clinical categories.

consistently rank lower across categories. Fine-tuning exhibits a model-dependent effect, with improvements observed in some cases but no consistent trend across all models. Performance is relatively higher in *Severity*, *Location*, and *Diagnostic*, and drops for *Reasoning* and *Treatment*.)*Deepali-edit*

G.3 ROUGE-L

(ROUGE-L shows moderate performance across categories, capturing partial structural overlap between predictions and references. General-purpose models consistently achieve the highest scores, with several fine-tuned variants among the top performers. Medical models demonstrate competitive performance, particularly in *Abnormality* and *Severity*, while proprietary models generally rank lower. The impact of fine-tuning is model-dependent, with improvements observed in some cases but no consistent trend across all models. Performance is stronger in *Severity*, *Location*, and *Diagnostic*, and declines for *Reasoning* and *Treatment*.)*Deepali-edit*

G.4 Semantic Similarity

(Semantic similarity scores are consistently high across all models and categories, indicating strong overall alignment in meaning between predictions and references. Differences between model groups are relatively small; however, general-purpose and medical models slightly outperform proprietary models in most categories. Fine-tuning shows minimal and model-dependent impact, with several fine-tuned models appearing among the top performers without a consistent trend. Performance remains uniformly strong across categories, with only slight reductions observed in *Treatment* and *Reasoning*.)*Deepali-edit*

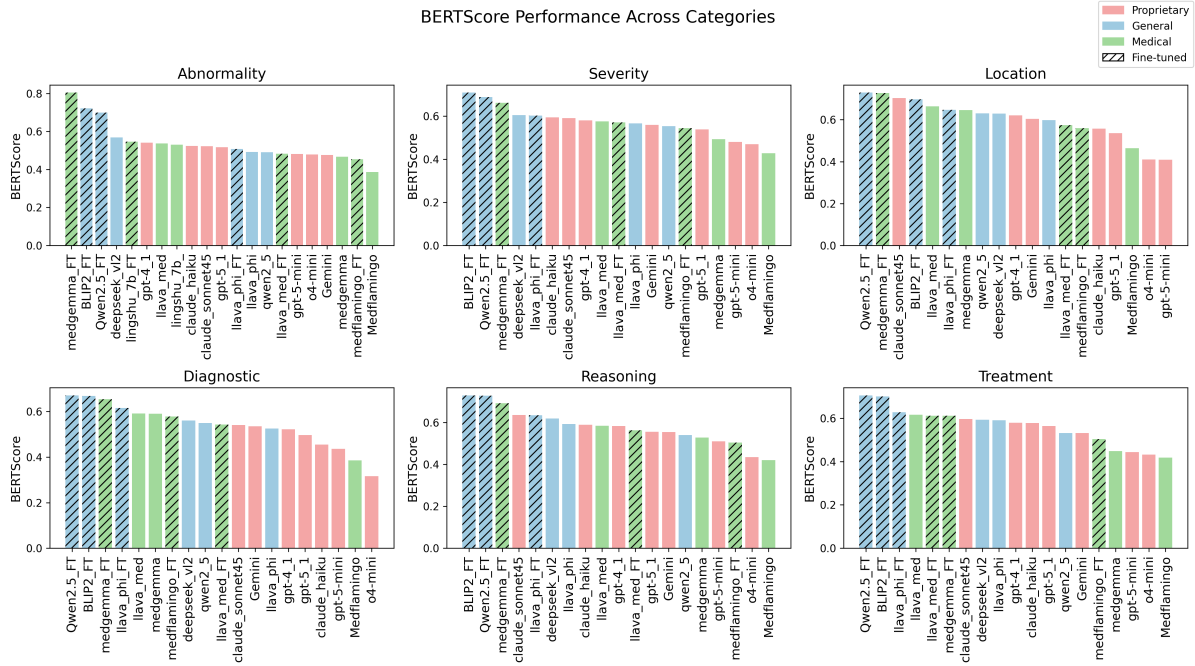


Fig. 5. BERTScore across clinical categories.

G.5 TF-IDF Similarity

(TF-IDF similarity shows moderate performance with clearer separation between model groups compared to embedding-based metrics. General-purpose and medical models consistently achieve higher scores across most categories, while proprietary models rank lower. Fine-tuned models frequently appear among the top performers, although the impact of fine-tuning remains model-dependent. Performance is relatively stronger in *Severity*, *Location*, and *Diagnostic*, and declines for *Reasoning* and *Treatment*, reflecting reduced term overlap in more complex responses.) *Deepali-edit*

H Model output error analysis

This section presents a fine-grained qualitative analysis of clinically relevant failure cases observed across the five representative models evaluated in the main text. For each model, we report selected question-answer examples covering abnormality detection, anatomical localization, diagnostic interpretation, and management reasoning, with clinically incorrect or misleading phrases explicitly highlighted. The examples reveal consistent failure patterns, including false-negative findings in the presence of fractures, substitution of acute traumatic pathology with chronic degenerative conditions, anatomically implausible localization across spinal regions, and unsafe or incomplete treatment recommendations. By directly contrasting model-generated responses with expert-validated ground truth, this analysis demonstrates how high semantic similarity scores can obscure clinically critical errors, underscoring the need for clinician-centered evaluation beyond surface-level text similarity in medical VQA systems.

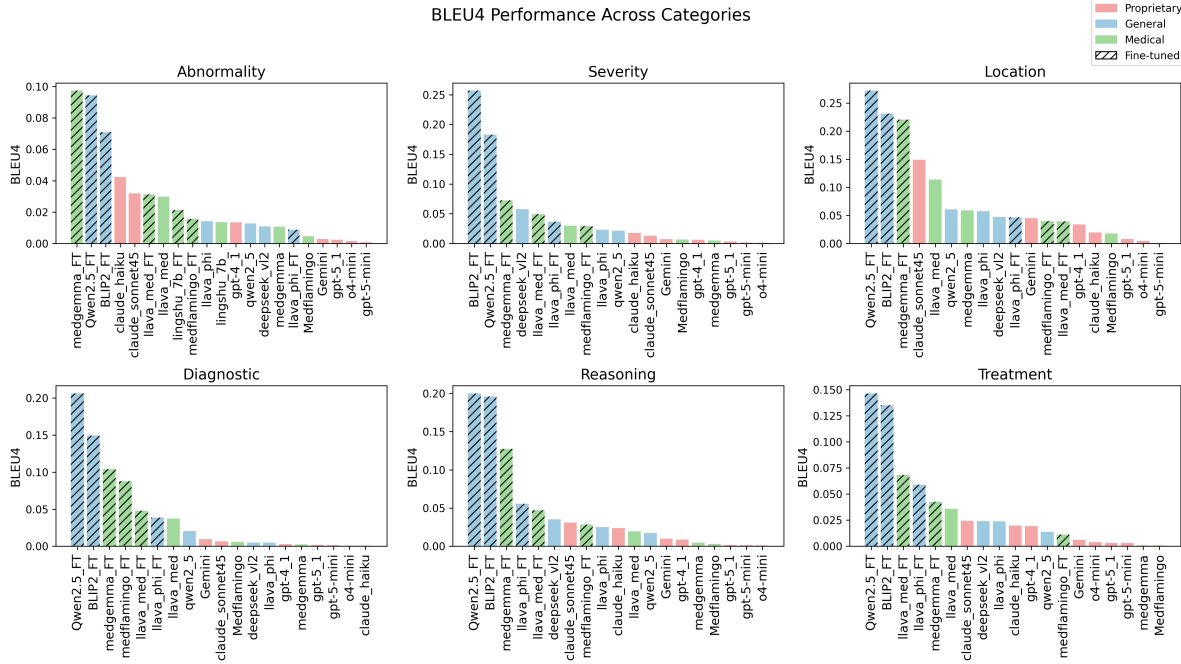


Fig. 6. BLEU-4 across clinical categories.

Table 6. Representative clinical failure cases across evaluated models. Clinically incorrect or misleading phrases are highlighted in **blue bold italics**.

Question	Ground Truth (Clinical)	Model Output	Error Type
BLIP-2			
Can you identify the radiographic abnormality present in the X-ray image?	Comminuted fracture of the C5 vertebra with vertebral body collapse and posterior element involvement.	The most likely clinical interpretation is <i>degenerative disc disease characterized by osteophytes and disc space narrowing</i> .	Pathology substitution
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are located at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Does this image reveal any pathological features?	Yes. The image shows a compression fracture of the C6 vertebra.	<i>No pathological features detected</i> .	False-negative detection

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Question	Ground Truth (Clinical)	Model Output	Error Type
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease with osteophyte formation</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the C6 vertebra in the cervical spine.	The anatomical site of the fracture is the <i>junction between C5 and C6</i> .	Anatomical inaccuracy
Does this image reveal any pathological features?	Yes. The image shows a fracture of the right clavicle.	<i>No pathological features detected.</i>	Missed extra-spinal pathology
Claude Sonnet			
Can you identify the radiographic abnormality present in the X-ray image?	Compression fracture of the C5 vertebra with vertebral body collapse.	The most likely clinical interpretation is <i>degenerative disc disease characterized by osteophytes</i> .	Pathology substitution
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are located at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Does this image reveal any pathological features?	Yes. The image shows a compression fracture of the C6 vertebra.	<i>No pathological features detected.</i>	False-negative detection
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease with osteophyte formation</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the C6 vertebra in the cervical spine.	The anatomical site of the fracture is the <i>junction between C5 and C6</i> .	Anatomical inaccuracy
Does this image reveal any pathological features?	Yes. The image shows a fracture of the right clavicle.	<i>No pathological features detected.</i>	Missed extra-spinal pathology
DeepSeek-VL2			
Can you identify the radiographic abnormality present in the X-ray image?	Compression fracture of the C6 vertebra with vertebral body collapse.	The image most likely represents <i>degenerative disc disease with osteophyte formation</i> .	Pathology substitution

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Question	Ground Truth (Clinical)	Model Output	Error Type
Does this image reveal any pathological features?	Yes. The image shows a comminuted fracture of the C5 vertebra.	<i>No pathological features detected.</i>	False-negative detection
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are present in the lumbar spine at the L1–L2 and L2–L3 levels.	The abnormalities are located at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar vertebral compression fracture with displacement.	The most likely clinical interpretation is <i>degenerative disc disease</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the L1 vertebra in the lumbar spine.	The fracture is located in the <i>cervical spine at C5–C6</i> .	Anatomical inaccuracy
Given the presence of detected abnormalities, what treatment or management strategies are recommended?	Immediate spinal immobilization with urgent surgical consultation.	<i>Conservative management with physiotherapy and lifestyle modification.</i>	Unsafe management suggestion
LingShu-7B (zero-shot)			
Can you identify the radiographic abnormality present in the X-ray image?	Comminuted fracture of the C5 vertebra with vertebral body collapse.	The findings are consistent with <i>degenerative disc disease with osteophyte formation</i> .	Pathology substitution
Does this image reveal any pathological features?	Yes. The image shows a compression fracture of the C6 vertebra.	<i>No pathological features detected.</i>	False-negative detection
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are present at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease</i> .	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the C6 vertebra in the cervical spine.	The fracture is located at the <i>junction between C5 and C6</i> .	Anatomical inaccuracy

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Question	Ground Truth (Clinician)	Model Output	Error Type
Given the presence of detected abnormalities, what treatment or management strategies are recommended?	Urgent spinal immobilization and surgical consultation are required.	<i>Conservative management with physiotherapy and pain control.</i>	Unsafe management recommendation
MedGemma (fine-tuned)			
Can you identify the radiographic abnormality present in the X-ray image?	Compression fracture of the C6 vertebra with loss of vertebral body height.	The findings are most consistent with <i>degenerative disc disease characterized by osteophytes.</i>	Pathology substitution
Does this image reveal any pathological features?	Yes. The image shows a fracture of the right clavicle.	<i>No pathological features detected.</i>	Missed extra-spinal pathology
Can you identify the specific areas within the image that show abnormalities?	Abnormalities are located in the lumbar spine at the L4–L5 and L5–S1 levels.	The abnormalities are identified at the <i>C5–C6 and C6–C7</i> levels.	Anatomical mislocalization
Given the X-ray image, what is the most probable clinical interpretation?	Lumbar spine fracture at L1 with vertebral body displacement.	The most likely clinical interpretation is <i>degenerative disc disease with disc space narrowing.</i>	Reasoning collapse
Can you identify the anatomical site of the fracture depicted in the image?	Fracture located at the L1 vertebra in the lumbar spine.	The fracture is located in the <i>cervical spine at C5–C6.</i>	Anatomical inaccuracy
Given the presence of detected abnormalities, what treatment or management strategies are recommended?	Immediate spinal immobilization with urgent surgical evaluation.	<i>Conservative treatment with physiotherapy and lifestyle modification.</i>	Unsafe management recommendation

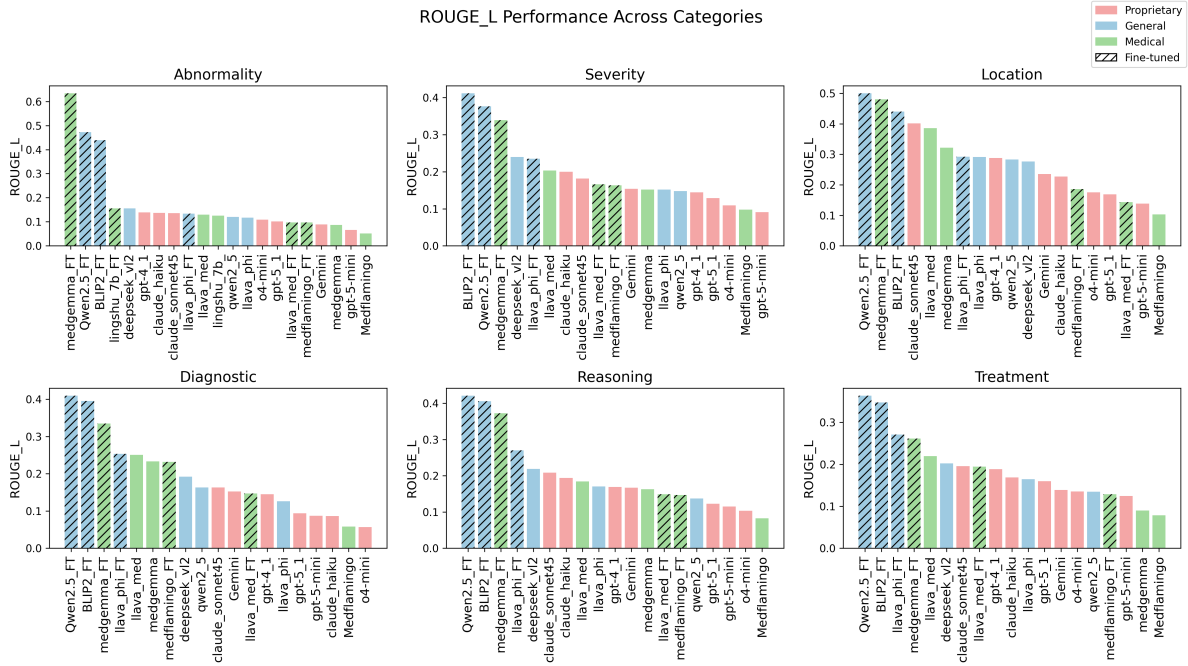


Fig. 7. ROUGE-L across clinical categories.

Table 2. Sample questions-answer pairs for Train set

Example 1: Image ID – vindr_train_4473.png		
Abnormality	Is there any abnormality present in this image?	No abnormality detected.
Example 2: Image ID – vindr_train_2027.png		
Abnormality	Is there any abnormality present in this image?	Yes, there is an abnormality present in this image. The X-ray shows a compression fracture of the L1 vertebra with evidence of callus formation, indicating a healing process.
Severity	What is the severity of the fracture?	The fracture is of moderate severity. It appears stable without significant displacement, although vertebral body compression may contribute to spinal stenosis.
Location	Where is the fracture located in the image?	The fracture is located at the L1 vertebra, just below the T12–L1 disc space, in the midline of the lumbar spine.
Diagnostic	What is the most likely diagnosis based on the X-ray image?	The most likely diagnosis is a compression fracture of the L1 vertebra, likely due to osteoporosis or trauma.
Reasoning	What are the possible causes of the fracture?	Possible causes include osteoporosis-related bone weakening or traumatic injury such as a fall or accident.
Treatment	What is the suggested treatment or management option for the detected abnormalities?	Conservative management including rest, pain control, and physical therapy is recommended, with surgical intervention considered in severe or unstable cases.
Example 3: Image ID – vindr_train_2530.png		
Abnormality	Is there any abnormality present in this image?	Yes, the image shows a compression fracture of the L1 vertebra with vertebral body height loss.
Severity	What is the severity of the fracture?	The fracture is of moderate severity and appears stable without rotational displacement.
Location	Where is the fracture located in the image?	The fracture involves the anterior aspect of the L1 vertebral body in the lumbar spine.
Diagnostic	What is the most likely diagnosis for the fracture?	A healing compression fracture of the L1 vertebra, potentially associated with osteoporosis or trauma.
Reasoning	What are the possible causes of the fracture?	Osteoporosis, traumatic injury, metabolic bone disease, or long-term medication use affecting bone density.
Treatment	What is the suggested treatment or management option for the fracture?	Conservative management including rest, analgesia, physical therapy, and possible bracing, with surgery reserved for unstable cases.

Table 3. Sample questions-answer pair for Test set

Example 1: Image ID – vindr_test_1251.png		
Abnormality	Does this image reveal any pathological features?	No abnormality detected.
Example 2: Image ID – vindr_test_538.png		
Abnormality	Can the X-ray image be evaluated for any radiographic abnormalities?	The X-ray image shows multiple abnormalities, including vertebral collapse, osteophyte formation, and the presence of a surgical implant.
Severity	Assess the degree of vertebral compression.	The vertebral collapse appears moderate to severe, with significant loss of vertebral height and potential compromise of spinal stability.
Location	Can you identify the specific areas within the image that show abnormalities?	The abnormalities are located in the lumbar spine, primarily at the L1–L2 and L2–L3 levels.
Diagnostic	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is degenerative disc disease with vertebral collapse and osteophyte formation, possibly complicated by prior surgical intervention.
Reasoning	Identify the underlying factors contributing to the observed abnormalities.	Potential contributing factors include degenerative disc disease, osteoporosis, prior trauma, or infection.
Treatment	Given the presence of detected abnormalities, what treatment or management strategies are typically recommended?	Recommended management includes conservative treatment with analgesics and physical therapy, surgical stabilization if indicated, and regular follow-up to monitor disease progression.
Example 3: Image ID – vindr_test_277.png		
Abnormality	Does this image reveal any pathological features?	Yes, the image demonstrates a compression fracture involving the C6 vertebra.
Severity	Assess the fracture's degree of displacement and comminution to determine its overall severity.	The fracture is of moderate severity, with partial compression and potential cervical spine instability but no complete displacement.
Location	Can you identify the anatomical site of the fracture depicted in the image?	The fracture is located at the C6 vertebra in the mid-cervical spine, approximately between the C5 and C7 levels.
Diagnostic	Given the X-ray image, what is the most probable clinical interpretation?	The most likely diagnosis is a compression fracture of the C6 vertebra.
Reasoning	Identify the underlying factors that may have contributed to the fracture.	Possible causes include traumatic injury, osteoporosis, or other conditions that weaken vertebral bone integrity.
Treatment	Given the presence of detected abnormalities, what are the recommended treatment or management strategies?	Management typically involves cervical spine immobilization using a collar or halo vest, followed by rehabilitation; surgical stabilization may be required in severe cases.

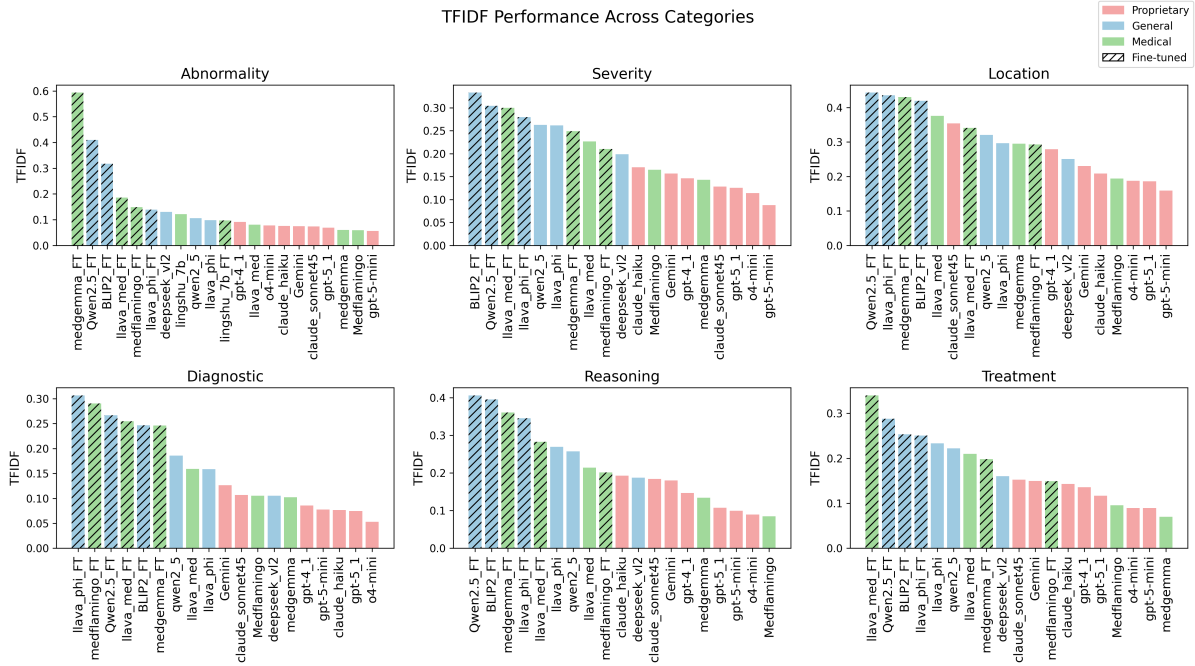


Fig. 8. TF-IDF similarity across clinical categories.